

Hai Bao

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Professional Experience

Sr. Data Scientist | Morningstar Sustainalytics | Jan 2023 - Mar 2026

- Streamlined data ingestion workflows to process 400,000+ news articles daily by engineering an automated ETL process using Python, which included data validation, reducing manual data handling and improving analysis turnaround.
- Optimized business support operations by integrating a GPT-powered LLM solution (OpenAI API), successfully reducing daily manual research tasks by 25% (1,000 articles) for the ESG Research team.
- Enhanced management's decision-making capabilities by engineering a Python-based Plotly dashboard (open source alternative to Power BI/Tableau) to track ESG Research workload distribution, enabling data-driven insights into team productivity and bottleneck identification.
- Helped Business Analysts with explaining AI model performance metric concepts such as Accuracy, Precision, Recall to the Senior Leadership Team.
- Led a team of 4 Data Scientists as part of the 'Digital Innovation' team for 6 months before a re-org moved the team onto the Data Science/AI team.

Data Scientist | Morningstar Sustainalytics | Jun 2022 - Jan 2023

- Optimized the analysis of public Sustainability/Annual Reports by developing an EU Taxonomy Detection tool, reducing manual review time by 25% through the implementation of a RAG-based data model to automate disclosure detection.
- Enhanced business analysis accuracy by reducing noise by 50% for the Thematic Crawler by engineering a multi-layer text mining framework to streamline data processing
- Input data from the Thematic Crawler was provided by stakeholders, so data cleaning/verification was performed on the input data for each request. This helped prevent erroneous data in the reports.

Business Analyst | Morningstar Sustainalytics | Nov 2021 - Jun 2022

- Orchestrated the deployment of customized Thematic Crawler applications using Python & SQL, resulting in a 75% reduction in manual data extraction time, by facilitating cross-functional requirement-gathering sessions to align technical capabilities with diverse business needs
- Delivered actionable insights via Excel reporting to support strategic decision-making for 5 business units

Technical Skills

- **Languages:** Python, VBA, SQL
- **Frameworks:** NumPy/Pandas, scikit-learn
- **Tools:** LLM, AWS, Docker, Git

Projects

Adventure Cycle Sales Dashboard

Power BI dashboard based on fictitious retail sales data with various KPIs.

Walmart Customer Segmentation

Using retail sales data provided by Walmart, the K-means Cluster Algorithm was applied to build separate customer profiles.

Rogers Centre Attendance Forecast

Loaded attendance data from MLB's StatsAPI into a Postgres database, then created a regression forecast using XGBoost of the attendance figures.

Education

- **B.S. in Statistics**, University of Toronto